VCF 9.x triage cheat sheet

Find the layer that owns the failure first, then open that layer's logs.

Which layer owns it?

	Management component	Core infrastructure
What	VCF Operations, VCF Automation, identity broker, Ops for networks, log management, services runtime	vCenter, ESX, NSX Managers and Edges, vSAN, HCX
Driven by	Fleet lifecycle (plans) and SDDC lifecycle (runs)	SDDC Manager
UI (9.1)	Build > Lifecycle > VCF Management > Tasks	Build > Lifecycle > VCF Instances
Errors	`VCF_LCM_…` codes with a reference ID	Enum codes with a reference token
Logs	Pods in `vcf-fleet-lcm`, `vcf-sddc-lcm`	`/var/log/vmware/vcf/` on SDDC Manager
Retry	Fix, then Retry Task (resumes at the failed step)	Fix, then UI retry or `PATCH /v1/tasks/{id}`

The method, in order

1. **Capture the IDs.** Reference ID or task ID, failed step, and the time to the minute with its time zone.
2. **Find the layer.** Management component or core infrastructure (table below).
3. **Read the message.** Including the resolution text in the task details, then the last "Caused by" in the log.
4. **Check for locks.** Nothing else running on the same scope: no parallel precheck, backup or rotation.
5. **Check the basics.** DNS (lowercase, forward and reverse), NTP, disk, reachability from the real source address.
6. **Recover in scope.** Retry Precheck, Retry Task, or the API. Never redeploy to retry.
7. **Escalate with evidence.** Bundle, IDs, timestamps, and what was tried since.

The reasoning behind each step: A seven-step triage for any VCF 9.x failure.

Error text to likely cause

`VCFMS-…`	Services runtime unhealthy. Fix pod health, then retry.
`VCFA-…`	VCF Automation topology or health validation.
`VCF_LCM_FLEET_UPGRADE_…`	Plan level. Then open the SDDC lifecycle task details.
`…IAM_401_INVALID_TOKEN…`	Time sync on every node in the cluster.
`"failed at step …"`	Fleet or SDDC lifecycle workflow. Follow the ID trail across services.
`UnknownHostException`	DNS.
`No route to host`	Network path or uplinks.
`InvalidLogin, 403, locked`	Credentials or lockout.
`PKIX path building failed`	Certificate trust.

Log map

SDDC Manager	`/var/log/vmware/vcf/`: `lcm/lcm-debug.log` (prechecks, bundles, upgrades), `domainmanager/` (domains, clusters, hosts, fleet task results), `operationsmanager/` (passwords, certificates, backups, commissioning), `commonsvcs/` (tokens, SSH keys, API), `sddc-support/` (SoS)
VCF Operations	`/storage/log/vcops/log/`: `collector.log`, `web-vcf.log` (Lifecycle plugins), `adapters/<Kind>/`, `casa/casa.log` (cluster)
Runtime (9.1)	Fleet build and upgrade in `vcf-fleet-lcm`, SDDC build and upgrade in `vcf-sddc-lcm`, depot in `vcf-fleet-depot`, platform tasks in `vmsp-platform`
vCenter	`/var/log/vmware/`: `vsphere-ui/logs/` (plugins), `cis-license/license.log`, `vmware-updatemgr/` (vLCM)
NSX	`/var/log/cm-inventory/cm-inventory.log` (compute managers), `/var/log/syslog`

Runtime toolkit (read-only)

```
# On a control-plane node: ssh vmware-system-user, then
sudo su -
export KUBECONFIG=/etc/kubernetes/admin.conf
```

```
kubectl get pods -A | grep -vE 'Running|Completed'
kubectl get tasks -n vmsp-platform --no-headers | grep -E '^(install|stage|configure)-component'
kubectl get events -n vcf-fleet-lcm --sort-by=.lastTimestamp | tail -30
P=$(kubectl get pods -n vcf-fleet-lcm -l app=fleetbuild -o jsonpath='{.items[0].metadata.name}')
kubectl logs $P -n vcf-fleet-lcm --tail=200
kubectl -n vcf-fleet-lcm exec vcf-fleet-lcm-db-0 -- patronictl list
```

Fixes go through KB scripts, not kubectl. More in the services runtime for VI admins.

Known fixes

9.1 binaries missing in VCF Operations 9.0	440282
vCenter not connected to a license server	440471
Import VCF Operations in fleet lifecycle fails	440459 · 441858
Failed or duplicate component	441333
Stale service account (InvalidLogin)	450375
SSH host key mismatch	316028
Stuck deployment lock	422426
Credential task holding a lock	444986
Stuck fleet or SDDC lifecycle task	435569
SoS runtime logs access_denied	451560
Depot 403 (9.0 token, 9.1 activation code)	420374 · 443647
Depot 403 from runtime nodes	443166
JWT token expired	409353

Evidence to attach

- **Core:** `sos --sddc-manager-logs --zip`, or `--collect-all-logs` (KB 385749)
- **Management services (9.1):** add `--vcf-mgmt-svcs-logs`, or the fleet lifecycle bundle. Fleet lifecycle down: KB 444466
- **VCF Operations:** Control Panel > Support Bundles (KB 443789)
- **UI problems:** a browser HAR file
- **Always:** reference and task IDs, time of failure, what was tried since

Rules: snapshot before any database change · one operation at a time · health before retry · retry before rebuild.

Upgrade order (9.0 to 9.1)

VCF Operations first, then vSphere Replication, Avi, SDDC Manager, then management services with the License Server and depot, identity broker, VCF Automation, then Orchestrator, Ops for networks and logs, then the core per domain (HCX, NSX Manager and vCenter, ESX, NSX Edges), then finalize (KB 440630).

Not official vendor guidance. Confirm against the current KB for your build before changing production.